

## LECTURE TWELVE

Model:

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2 \text{ReLU}(t-2) + \dots + \beta_{n-1} \text{ReLU}(t-(n-1)) + \varepsilon_t$$

$t = 1, \dots, n$

→ high-dimension linear regression

$$\boxed{y = X\beta + \varepsilon} \quad X = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ \vdots & 1 & 0 & \dots & 0 \\ 0 & 1 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n-1 & n-2 & \dots & 1 \end{bmatrix}$$

$n \times n$

① Least Squares → Overfitting  
(fitted values = data)

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$X\hat{\beta} = y \rightarrow \text{overfitting}$$

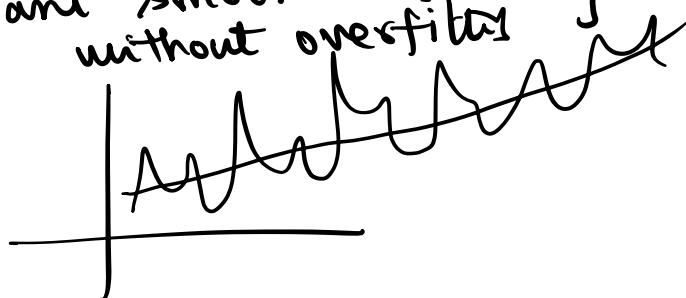
② Regularization.

Ridge:  $\|y - X\beta\|^2 + \lambda \left[ \sum_{j=2}^{n-1} \beta_j^2 \right]$

LASSO:  $\|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} |\beta_j|$

Why Regularize?

① want smoother fits without overfitting } Our preference



② Regularization leads to improved prediction.  
 (underlies CV for  $\lambda$ -selection)  
 Cross Validation

---

Formula for the Ridge Estimator

$$\|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2$$

$$(y - X\beta)^T (y - X\beta)$$

$$\nabla_{\beta} \left[ \underbrace{\beta^T X^T X \beta}_{\text{}} - \underbrace{2\beta^T X^T y}_{\text{}} + \underbrace{y^T y}_{\text{}} + \lambda \sum_{j=2}^{n-1} \beta_j^2 \right]$$

$$= \cancel{2} \underbrace{X^T X \beta}_{\text{}} - \cancel{2} X^T y + \cancel{2\lambda} \begin{pmatrix} 0 \\ 0 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \end{pmatrix} = 0$$

$$\begin{pmatrix} 0 \\ 0 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \end{pmatrix} = J\beta \text{ where}$$

$$J = \begin{bmatrix} 0 & & & 0 \\ & 0 & & \\ & & 1 & \\ 0 & & & \ddots \\ & & & & 1 \end{bmatrix}$$

$$X^T X \beta - X^T y + \lambda J \beta = 0$$

$$(X^T X + \lambda J) \beta = X^T y$$


---

$$\hat{\beta}^{\text{Ridge}}(\lambda) = (X^T X + \lambda I)^{-1} X^T y$$

$$\hat{\beta}^{\text{least square}} = (X^T X)^{-1} X^T y \quad \leftarrow \lambda=0$$

## Bayesian Regularization

$$y = X\beta + \varepsilon, \quad \varepsilon_i \text{ iid } N(0, \sigma^2)$$

$$\text{likelihood: } \left( \frac{1}{\sqrt{2\pi} \sigma} \right)^n \exp \left[ -\frac{\|y - X\beta\|^2}{2\sigma^2} \right]$$

$$\beta (\beta_0, \beta_1, \dots, \beta_{n-1}) \propto \sigma$$

$$\textcircled{1} \beta_0, \beta_1, \dots, \beta_{n-1} \text{ iid } \underset{C \rightarrow \infty}{\sim} \text{Unif}(-C, C)$$

↓ when  $C \rightarrow \infty$

$$\text{posterior } \beta \mid \text{data } \underset{\sigma}{\sim} N \left( (X^T X)^{-1} X^T y, \sigma^2 (X^T X)^{-1} \right)$$

↓  
n-variate normal

Homework 1

$$\textcircled{2} \beta_0, \beta_1, \dots, \beta_{n-1} \text{ iid } \underset{(C \text{ large})}{\sim} N(0, C)$$

↓ posterior

$$\beta \mid \text{data } \underset{\sigma}{\sim} \frac{1}{\sqrt{2\pi} C} \exp \left[ -\frac{\beta^2}{2C} \right]$$

behaves like a constant

$$\sim N\left(\left(\frac{X^T X}{\sigma^2} + \frac{I}{C}\right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + \frac{I}{C}\right)^{-1}\right)$$

Valid for every  $C$   
(not just for large  $C$ )

for most  $\beta_j$   
very similar to  $\text{Unif}(-C, C)$   
 $\frac{I\{-C < \beta_j < C\}}{2C}$

If  $C \rightarrow \infty$ :  $N\left(\left(\frac{X^T X}{\sigma^2}\right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2}\right)^{-1}\right)$   
 $= N\left((X^T X)^{-1} X^T y, \sigma^2 (X^T X)^{-1}\right)$

coincides with posterior for  $\text{Unif}(-\infty, \infty)$  prior.

$$\frac{1}{\sqrt{2\pi} C} \exp\left(-\frac{\beta_j^2}{2C}\right)$$

$$C = 10^{10}$$

$$\beta_j \in (-10, 10)$$

### Summarize

① If you are using  $C$  large (either in  $\text{Unif}(-C, C)$  or  $N(0, C)$ ) the posterior mean = least squares will lead to over-fitting.

③  $\beta_0, \beta_1 \stackrel{\text{iid}}{\sim} N(0, C)$ ,  $\beta_2, \dots, \beta_{n-1} \stackrel{\text{iid}}{\sim} N(0, \tau^2)$   
 $C$ : large for some small  $\tau > 0$

This will lead to smooth fits without overfitting (depending on the chosen value of  $\tau$ )

The formula for the posterior:

$$\beta \sim N_n \left( \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \underbrace{\begin{pmatrix} C & & 0 \\ & C\tau^2 & \\ 0 & \ddots & \tau^2 \end{pmatrix}}_Q \right)$$

$$\beta \sim N(0, Q)$$

Check: posterior now becomes

$$N \left( \left( \frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2}, \left( \frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \right)$$

$$\beta | \text{data}_\sigma$$

posterior mean:

$$\left( \frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2}$$

$$Q^{-1} = \begin{pmatrix} 1/C & & 0 \\ & 1/C & \\ & & 1/\tau^2 & \\ 0 & & & \ddots & 1/\tau^2 \end{pmatrix}$$

$$\approx \begin{pmatrix} 0 & 0 & 1/\tau^2 & 0 \\ & 0 & & 1/\tau^2 \end{pmatrix} = \frac{1}{\tau^2} J$$

posterior mean:  $\left( \frac{X^T X}{\sigma^2} + \frac{1}{\tau^2} J \right)^{-1} \frac{X^T y}{\sigma^2} = \left( X^T X + \frac{\sigma^2}{\tau^2} J \right)^{-1} X^T y$

Ridge:  $(X^T X + \lambda J)^{-1} X^T y$

Coincide if  $\lambda = \frac{\sigma^2}{\tau^2}$

$\beta_j \stackrel{iid}{\sim} N(0, c)$  (or  $Unif(-c, c)$ )  
 $c$  large

Bayesian = Frequentist  
 $(X^T X)^{-1} X^T y$

$\beta_0, \beta_1 \stackrel{iid}{\sim} N(0, c)$ ,  $\beta_j \stackrel{iid}{\sim} N(0, \tau^2)$   
 $c$  large,  $j=2, \dots, n-1$

Frequentist  
 with Ridge  
 Regularization = Bayesian

Prior 1  $\rightarrow \beta_j \stackrel{iid}{\sim} N(0, c) \rightarrow$  least square

Prior 2  $\rightarrow \beta_0, \beta_1 \stackrel{iid}{\sim} N(0, c)$ ,  $\beta_j \stackrel{iid}{\sim} N(0, \tau^2) \rightarrow$  Ridge  
 Regularization  
 if  $\tau$  is small

Prior 2 is restrictive.

Prior 3:  $\beta_0, \beta_1 \stackrel{iid}{\sim} N(0, C), \beta_j \stackrel{iid}{\sim} N(0, \tau^2)$   
 $\log \tau \sim \text{Unif}(-C, C), \log \sigma \sim \text{Unif}(\epsilon, \infty)$

More General

prior:

$$f(\beta, \sigma, \tau) \propto \frac{1}{\sqrt{\det Q}} \exp\left(-\frac{\beta^T Q^{-1} \beta}{2}\right) \frac{I\{-C < \log \tau < C\}}{\tau} \frac{I\{-C < \log \sigma < \infty\}}{\sigma}$$

$$\beta \sim N\left(\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} C & & & \\ & C & & \\ & & \tau^2 & \\ & & & \ddots & \\ 0 & & & & \tau^2 \end{pmatrix}\right) = \frac{\tau^{-1} \sigma^{-1}}{\sqrt{\det Q}} \exp\left(-\frac{\beta^T Q^{-1} \beta}{2}\right)$$

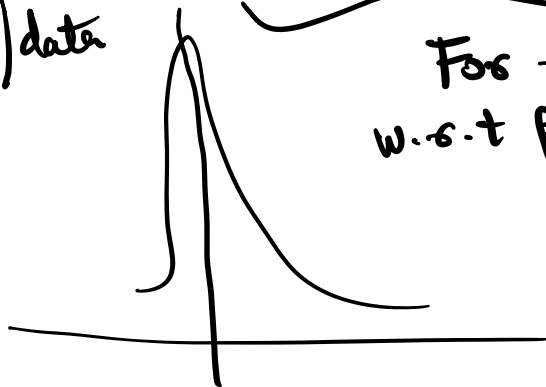
$N(0, Q)$

likelihood:  $\sigma^{-n} \exp\left(-\frac{\|y - X\beta\|^2}{2\sigma^2}\right)$

posteriors  $(\beta, \tau, \sigma)$ :  $\frac{\tau^{-n-1}}{\sqrt{\det Q}} \exp\left[-\frac{1}{2} \left( \frac{\|y - X\beta\|^2}{\sigma^2} + \beta^T Q^{-1} \beta \right)\right]$

$f(\beta, \tau, \sigma)$   
 $\beta, \tau, \sigma | \text{data}$

For fixed  $\sigma, \tau$ , integration w.r.t  $\beta$  is tractable.



$$\frac{\|y - X\beta\|^2}{\sigma^2} + \beta^T Q^{-1} \beta$$

$$x^2 - 4x + 6 = (x-2)^2 + 2$$

$$= (\beta - b)^T \left( \frac{X^T X}{\sigma^2} + Q^{-1} \right) (\beta - b) + \frac{y^T y}{\sigma^2} - b^T \left( \frac{X^T X}{\sigma^2} + Q^{-1} \right) b$$

$$b = \left( \frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2}$$

$f(\sigma, \tau)$  is integral of  $f(\beta, \sigma, \tau)$  w.r.t  $\beta$   
 $\sigma, \tau | \text{data}$

$$\propto \frac{\sigma^{-n-1} \tau^{-1}}{\sqrt{\det Q}} \sqrt{\det \left( \frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1}} \exp \left( -\frac{y^T y}{2\sigma^2} \right) \exp \left( \frac{y^T X \left( \frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2}}{2\sigma^2} \right)$$

- ① Take a grid of values of  $\sigma$  &  $\tau$
- ② Compute  $f(\sigma, \tau | \text{data})$  over the grid
- ③ posterior mean/mode of  $\sigma$  &  $\tau$   
 Sample values of  $\sigma$  &  $\tau$
- ④ Sample  $\beta$  from  $\beta | \text{data}, \sigma, \tau$  : normal posterior

Why no overfitting?

$f_{\text{data} | \beta, \sigma}$  : likelihood maximization leads to  $\hat{\beta} = \text{unregularized least squares}$   $\hat{\sigma} = 0$

$$f_{\text{data} | \tau, \sigma} = \int f_{\text{data} | \beta, \sigma}^{(\text{data})} f_{\beta | \tau}(\beta) d\beta$$

$f_{\tau, \sigma | \text{data}}$

$$f_{\tau, \sigma | \text{data}} \propto \boxed{f_{\text{data} | \tau, \sigma}} \underbrace{f_{\tau, \sigma}}$$

$$\updownarrow$$

$$\boxed{f_{\text{data} | \beta, \sigma}}$$

↑